Set 1:

Group A (Answer TWO questions. 2x12=24)

1. What are the different models for deployment in cloud computing? Explain any one of them. Discuss Infrastructure as a Service (IaaS) vs Software as a Service (SaaS) vs Platform as a Service (PaaS). (6+6)
2. What is virtualization and what are its benefits? Briefly explain how hardware resources such as processors, storage, and networks can be virtualized. (5+7)
3. What is Microsoft Azure? Define Azure services and explain them with the help of suitable examples. (2+10)

Group B (Answer SIX questions. 6x6=36)

1. What do you mean by cloud and cloud computing? Give a brief note on the advantages and disadvantages of cloud computing.
2. Public cloud is less secure. Justify.
3. What are the various cloud security approaches and cloud security challenges? Discuss briefly.
4. What is Service Level Agreement (SLA)? Explain data stores in the cloud.
5. What is the role of hypervisor in virtualization? Briefly explain the different types of hypervisors with a neat diagram.
6. Discuss the authentication process in cloud computing. Mention the requirements for client access in the cloud.
7. Write short notes on any TWO:
    - Amazon EC2
    - Private Cloud
    - Big Data Analysis

Set 2:

Group A (Answer TWO questions. 2x12=24)

1. Explain the evolution of cloud computing. What are the benefits and limitations of cloud computing? (6+6)
2. Describe the various cloud computing service models. Discuss the key characteristics and applications of SaaS, IaaS, and PaaS. (6+6)
3. What is Amazon Web Services (AWS)? Explain the installation and configuration process of AWS. Discuss its main features. (4+8)

Group B (Answer SIX questions. 6x6=36)

1. Define cloud computing. How does it differ from traditional computing?
2. Explain the concept of a hybrid cloud. What are its advantages and disadvantages?
3. Describe the objectives and benefits of virtualization. How does virtualization improve the efficiency of cloud computing?
4. What is the significance of data storage virtualization in cloud computing?
5. Discuss the challenges and solutions for cloud security. How can encryption and tokenization enhance cloud security?
6. Explain the role of SOA (Service-Oriented Architecture) in cloud computing. How does it improve performance through load balancing?
7. Write short notes on any TWO:
    - Google App Engine
    - Cloud-based Big Data/Real-time Analytics
    - Commercial and Business Considerations in Cloud Computing

Set 3:

Group A (Answer TWO questions. 2x12=24)

1. What are cloud deployment models? Describe private, public, and hybrid clouds. Discuss their advantages and disadvantages. (6+6)
2. Explain the concept of virtualization. What are the differences between emulation and virtualization? (6+6)
3. What are Service Level Agreements (SLAs) in cloud computing? Discuss their importance and key components. (4+8)

Group B (Answer SIX questions. 6x6=36)

1. Define cloud architecture. How does it support scalability and reliability in cloud services?
2. Compare and contrast cloud computing with grid computing.
3. What are the economic benefits of cloud computing for businesses?
4. Discuss the process and benefits of deploying a web service from inside and outside of a cloud.
5. What are the common security challenges faced by cloud computing? How can they be mitigated?
6. Explain the role of VMware in server virtualization.
7. Write short notes on any TWO:
    - Eucalyptus
    - Cloud security models and related patterns
    - Database and data stores in the cloud

Set 4:

Group A (Answer TWO questions. 2x12=24)

1. Discuss the different types of cloud computing (private, public, hybrid). Explain the characteristics of each. (6+6)
2. What is the significance of cloud-based big data/real-time analytics? How does it benefit businesses? (6+6)
3. Describe the cloud service models SaaS, IaaS, and PaaS. Provide examples of each and explain their use cases. (6+6)

Group B (Answer SIX questions. 6x6=36)

1. What are the key benefits and limitations of cloud computing? Provide examples.
2. Explain the role of load balancing in cloud computing. How does it improve performance?
3. Discuss cloud security alliance standards. How do they help in securing cloud environments?
4. Describe the main features and benefits of Amazon EC2.
5. What is cloud contracting? Discuss its commercial and business considerations.
6. Explain the importance of data scalability in cloud services.
7. Write short notes on any TWO:
    - Mainstream cloud security offerings
    - Virtualization for enterprise
    - Installation of AWS

Set 5:

Group A (Answer TWO questions. 2x12=24)

1. What is cloud architecture? Explain its components and how it supports cloud services. (6+6)
2. Describe the process of deploying a web service on a cloud platform. What are the key considerations? (6+6)
3. Explain the concept of data storage virtualization. How does it benefit cloud computing? (6+6)

Group B (Answer SIX questions. 6x6=36)

1. Define cluster computing and grid computing. How do they compare to cloud computing?
2. Discuss the role of hypervisors in virtualization. Provide examples of different types of hypervisors.
3. Explain the significance of understanding SOA in cloud computing.
4. What are the objectives and benefits of using VMware in server virtualization?
5. Describe the authentication process in cloud computing. What are the requirements for client access?
6. Discuss the role of Google App Engine in cloud computing. What are its main features?
7. Write short notes on any TWO:
   - Cloud computing benefits for SMEs
   - Storage as a Service (StaaS)
   - Improving performance through load balancing